Sharp growth of corner cuts in fixed dimension

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Working mathematical draft. The argument has passed an internal proof audit; the priority assessment and submission review remain incomplete.

Abstract

Let $C_d(n)$ count the n-point subsets of the nonnegative integer lattice that can be strictly separated from their complements by an affine hyperplane. For every fixed $d \geq 2$, we prove

$$C_d(n) = \Theta_d\big(n^{d-1}(\log n)^{d-1}\big).$$

The upper bound is due to Wagner. For the lower bound, a finite sieve produces primitive lattice simplices in separated axis boxes. Their primitive normal directions are distinguished at a prescribed cardinality by two integer patches on opposite sides of a boundary level. Summing over disjoint aspect ranges supplies the logarithmic factor. This gives a matching lower bound in every fixed dimension, including the three-dimensional order $n^2(\log n)^2$. The cumulative count is consequently of order $n^d(\log n)^{d-1}$.

1 The counting problem and its scope

A nonempty finite set $D \subset \mathbb{Z}^d_{\geq 0}$ is a *corner cut* if

$$D = \{x \in \mathbb{Z}^d_{\geq 0} : w \cdot x < t\} \quad \text{for some } w \in \mathbb{R}^d_{>0},\ t > 0,$$

with no lattice point on the cutting hyperplane. Write $C_d(n)$ for the number of such sets of cardinality n, and put

$$A_d(n) = \sum_{j=1}^{n} C_d(j).$$

We count sets, rather than the orders of their elements or the weights representing them. Every cut is an order ideal for the coordinate order. In particular, every coordinate of an element of an n-point cut is at most $n-1$, so these counts are finite.

These objects occur as the exponent sets of initial segments of smooth numbers. For distinct primes $p_1, \ldots, p_d$, comparing the products $\prod_i p_i^{x_i}$ is equivalent to comparing $\sum_i x_i \log p_i$. An exponent prefix is consequently a corner cut. Conversely, every corner cut has such a realization, even with all primes above any prescribed bound (Proposition 6). Thus the full geometric class is exactly the class of exponent-prefix sets as the ordered prime bases vary. This does not give a prime-distribution estimate or a faster random-access algorithm for a fixed prime set.

Corner cuts were introduced in connection with state polyhedra and Gröbner bases by Onn and Sturmfels [1]. The planar order $C_2(n) = \Theta(n \log n)$ is due to Corteel, Rémond, Schaeffer and Thomas [2]. For fixed d, Wagner [3] proved

$$\Omega_d(n^{d-1} \log n) \leq C_d(n) \leq O_d(n^{d-1}(\log n)^{d-1}). \tag{1}$$

His thesis [4, Chapter 3] gives the lower and upper arguments and suggests seeking greater completion multiplicity. We obtain the lower bound in the same dimension: we count primitive normals and show that two of them cannot define the same sufficiently large prefix. No monotonicity assumption or inference from an average is used.

Theorem 1. *For each fixed integer $d \geq 2$,*

$$C_d(n) = \Theta_d\big(n^{d-1}(\log n)^{d-1}\big).$$

Here is one explicit lower bound. Put

$$K_d = 4^{d+1}, \qquad T_d = [64d^2(K_d+1)]^d, \qquad R_0(d) = 4d^2(d+1),$$

$$n_*(d) = T_d 4^{R_0(d)}, \qquad \rho_d = 8^{-(d-1)}(100d)^{-d^2},$$

and

$$\lambda_d = \frac{\rho_d}{2^d(4T_d)^{d-1}[4(d+1)]^{d-1}(d-1)!}.$$

For every integer $n \geq n_*(d)$,

$$C_d(n) \geq \lambda_d n^{d-1}(\log_4 n)^{d-1}. \tag{2}$$

Summing gives $A_d(n) = \Theta_d(n^d(\log n)^{d-1})$. The pointwise conclusion matches (1) and strengthens its lower logarithmic exponent when $d \geq 3$. This comparison is with the inspected baseline; the priority status is discussed at the end of the paper.

2 Primitive simplices in axis boxes

Fix an ambient integer $d \geq 2$ and set $\delta = 1/(100d)$. For integer side lengths $L_1, \ldots, L_d$, put

$$m = \min_i L_i, \quad M = \max_i L_i, \quad Q = \prod_{i=1}^{d} L_i.$$

Define $B_i \subset \mathbb{Z}^d_{\geq 0}$ by taking the jth coordinate interval to be

$$[L_j, L_j + \lfloor \delta L_j \rfloor] \quad (j = i), \qquad [0, \lfloor \delta L_j \rfloor] \quad (j \neq i).$$

A d-by-$(d-1)$ integer matrix is called primitive if the greatest common divisor of its maximal minors is one.

Lemma 2. *Assume $M \leq m^2$ and $m \geq 2^{8d^2}$. At least a fraction $8^{-(d-1)}$ of the ordered choices $p_i \in B_i$ have primitive difference matrix*

$$V = [p_2 - p_1 \ \cdots \ p_d - p_1].$$

The minor criterion and its equivalence with full column rank modulo every prime are standard [8, Propositions 3–4]. The point requiring proof here is uniformity for these moving, possibly anisotropic boxes.

Proof. Keep the same δ throughout a dimension induction. Let E_r be the event that the first r points, projected onto the first r coordinates, have primitive difference matrix. The event E_1 is automatic, with empty minor equal to one. At step $2 \leq r \leq d$, fix a primitive configuration of the previous $r-1$ projected points. Its difference matrix U has size $(r-1)$-by-$(r-2)$. Its vector a_0 of signed maximal minors is primitive, and U has rank $r-2$ modulo every prime.

Let u be the first $r-1$ coordinates of $p_r - p_1$, and put $D = \det[U\ u]$. This determinant is nonzero for every allowed choice. Indeed, the normalized row matrix of the previous $r-1$ projected points differs from the identity by at most $(r-1)\delta$ in maximum row-sum norm. The normal of their affine span, normalized to height one, has scaled coordinates within $(r-1)\delta/(1-(r-1)\delta)$ of one and is positive. The new projected point has every scaled coordinate at most δ; it is strictly on the origin side of that affine span. Also,

$$|D| \le (r-1)!\, 2^{r-1} M^{r-1} \le 2^{d^2} m^{2(d-1)}. \tag{3}$$

Every coordinate interval contains more than δm integers. For a prime ℓ, the equation $D \equiv 0 \pmod{\ell}$ is a nontrivial linear equation in the new projected point: at least one coordinate of a_0 is nonzero modulo ℓ. Its probability is at most $1/\ell + a/m$, where $a = 100d$.

Suppose $\ell \mid D$. Since U has full column rank modulo ℓ, there is a unique vector λ with $u \equiv U\lambda \pmod{\ell}$. Write z for the rth row of the old columns and u_r for the rth coordinate of the new column. The full r-by-$(r-1)$ matrix loses rank modulo ℓ precisely if

$$u_r \equiv z\lambda \pmod{\ell}. \tag{4}$$

Conditional on everything except the last coordinate of p_r, this is one residue class, of probability at most $1/\ell + a/m$. Thus the probability of failure at any particular prime is at most $(1/\ell + a/m)^2$.

Put $s = \lfloor (\log_2 m)/2 \rfloor$, $P = 2^s$. The small-prime union bound gives

$$\begin{aligned} \Pr(\text{failure at a prime } \ell \le P) &\le \sum_{\substack{\ell \le P \\ \ell \text{ prime}}} (1/\ell + a/m)^2 \\ &\le \frac{3}{4} + \frac{2a(s+1)}{2^{2s}} + \frac{a^2}{2^{3s}}. \end{aligned} \tag{5}$$

Here $\sum_{j=2}^{\infty} j^{-2} < 3/4$ follows by comparison with $1/4 + \int_2^{\infty} x^{-2}\,dx$. The cross term uses $H_P \le 1 + \log P \le 1 + s$, and the constant term uses at most P primes.

For large primes, first fix the new projected coordinates. Every bad prime $\ell > P$ divides the same nonzero integer D. By (3), their number is at most

$$\frac{\log_2 |D|}{s} \le 4(d-1) + \frac{d^2 + 4(d-1)}{s} \le b := 4d - 3 \quad \text{if } s \ge d^2 + 4(d-1).$$

For each of these finitely many primes, (4) bounds the conditional last-coordinate failure probability by $2^{-s} + a2^{-2s}$. Their total contribution is therefore at most $b(2^{-s} + a2^{-2s})$.

Define

$$E_d(s) = \frac{2a(s+1)}{2^{2s}} + \frac{a^2}{2^{3s}} + b(2^{-s} + a2^{-2s}). \tag{6}$$

This decreases for $s \ge 1$. For every integer $d \ge 2$, the choice $S_d = 4d^2$ satisfies $S_d \ge d^2 + 4(d-1)$ and $E_d(S_d) < 2^{-10} < 1/8$. For an explicit verification valid in every dimension, use $a \le 2^{4d}$, $b \le 2^{2d}$ and $4d^2 + 1 \le 2^{3d}$. They give

$$E_d(4d^2) \le 2^{1+7d-8d^2} + 2^{8d-12d^2} + 2^{2d-4d^2} + 2^{6d-8d^2} < 2^{-10}.$$

The four exponents decrease for $d \ge 2$, and at $d = 2$ are $-17, -32, -12, -20$.

Consequently the conditional probability of a nonprimitive extension is at most $3/4 + E_d(s) \le 7/8$. To justify the conditioning across steps, the events $E_1, \dots, E_{r-1}$ depend only on previous projected coordinates. After these are fixed, the old rth coordinates and the new point retain their product distributions. Thus

$$\Pr(E_r \mid E_1, \dots, E_{r-1}) \ge 1/8.$$

Multiplication proves that $\Pr(E_1 \cap \cdots \cap E_d) \ge 8^{-(d-1)}$. This intersection implies E_d, as required. All bounds on large primes were finite before any limit was taken. $\square$

3 Primitive normals at a prescribed size

For each primitive frame from Lemma 2, orient its cofactor normal a positively and put $H_a = a \cdot p_1$. Let P have rows p_i. Its column-scaled matrix differs from the identity by a matrix of maximum row-sum norm at most $d\delta = 1/100$. The determinant is positive and every eigenvalue lies within $1/100$ of one. Consequently

$$H_a = \det P, \qquad 2^{-d}Q \le H_a \le 2^d Q. \tag{7}$$

The solution of $p_i \cdot w = 1$ satisfies $|L_i w_i - 1| \le d\delta/(1 - d\delta) < 1/10$. Since $w = a/H_a$, this gives

$$\frac{9}{10} < \frac{a_i L_i}{H_a} < \frac{11}{10}. \tag{8}$$

Lemma 3. *Under the hypotheses of Lemma 2, these frames give at least $2^{-d}\rho_d Q^{d-1}$ distinct primitive positive normals.*

Proof. Primitivity gives an integer z with $a \cdot z = 1$. The matrix $[V\ z]$ is unimodular, so the columns of V form a basis of the entire integer kernel of a. If two points of the same B_i lie in the selected affine plane, their difference is Vc, with $c \in \mathbb{Z}^{d-1}$. The submatrix in rows $2, \dots, d$, after row scaling by the L_i, differs from the identity by at most $(d-1)\delta$ in maximum row-sum norm. Thus

$$\|c\|_\infty \le \frac{\delta}{1 - (d-1)\delta} < 1,$$

so the two points agree. Each plane determines its selected simplex. Each box has at least $\delta^d Q$ points, so the number of distinct planes is at least $\rho_d Q^d$. For any fixed primitive normal, the integer height in (7) has at most $2^d Q$ possibilities. Division proves the claim. □

When one normal occurs at several heights, fix one witnessing simplex for it, and use that simplex to define H_a. The prefix below depends only on a, not on this choice of witness.

For a positive integer normal a, order lattice points by $(a \cdot x, x_1, \dots, x_d)$, and let $D_n(a)$ be the first n. Let h be the least integer for which

$$F_a(h) := |\{x \in \mathbb{Z}^d_{\ge 0} : a \cdot x \le h\}| \ge n.$$

The set $D_n(a)$ is a corner cut, including when the score-h level is only partly included. Indeed $h \le (n-1)\min_i a_i$, by considering a minimum-weight axis. Every point at score at most h lies in $\{0, \dots, n-1\}^d$. For $n > 1$, the integer vector $u_i = n^{d-i}$ represents lexicographic order in that box, and $u \cdot x \le n^d - 1$ there. Taking $w = n^d a + u$ preserves all these levels and strictly orders the boundary level. Positivity keeps every higher a-level above them. A threshold between consecutive integer scores gives a strict representation. The case $n = 1$ is immediate. We will use the weak separation property

$$b \cdot x \le b \cdot y \quad (x \in D_n(b),\ y \notin D_n(b)). \tag{9}$$

Lemma 4 (Fixed-size injectivity)**.** *Let a, b be primitive positive normals furnished by the same aspect box. Put $K = 4^{d+1}$. If the boundary height h of $D_n(a)$ satisfies*

$$h/H_a \ge 16d^2(K+1), \tag{10}$$

then $D_n(a) = D_n(b)$ implies $a = b$.

Proof. Choose the selected frame $V = [v_1 \ \cdots \ v_{d-1}]$ for a. Since V is a basis of its integer kernel, an integer solution of $a \cdot z = 1$ can be reduced modulo V to give

$$z = p_1/H_a + Vc, \qquad |c_j| \le 1/2. \tag{11}$$

Each entry satisfies $|(v_j)_i| < 2L_i$, so $|z_i| \le (d+1)L_i \le 2dL_i$. Let $y_i = (h-1)/(da_i)$. Reduce $(h-1)z$ modulo V about y, obtaining an integer x_0 such that

$$a \cdot x_0 = h - 1, \qquad x_0 - y = V\beta, \qquad |\beta_j| \le 1/2.$$

Both integer patches

$$\{x_0 + Vk : k \in \mathbb{Z}^{d-1},\ |k_j| \le K\}, \qquad \{x_0 + 2z + Vk : k \in \mathbb{Z}^{d-1},\ |k_j| \le K\} \tag{12}$$

are nonnegative. To see this coordinatewise, the displacement from y_i is less than $(2Kd + 5d)L_i$, while (8) and (10) give

$$y_i/L_i > \frac{h-1}{2dH_a} > 2Kd + 5d.$$

Their a-scores are respectively $h-1$ and $h+1$. Thus the first patch is entirely inside $D_n(a)$, the second entirely outside; the score-h tie is irrelevant.

Assume the two prefixes agree, and put $S = \sum_j |b \cdot v_j|$. By (9), the maximum b-score on the first patch cannot exceed the minimum on the second. Hence

$$KS \le b \cdot z \le b \cdot p_1/H_a + S/2. \tag{13}$$

For two normals in the same aspect family, (7) gives $H_b/H_a \le 4^d$, and (8) gives

$$\frac{b \cdot p_1}{H_a} < 4^d \frac{11}{10}(1 + d\delta) < 2 \cdot 4^d.$$

It follows that $(K - 1/2)S < 2 \cdot 4^d < K - 1/2$. But S is an integer, so $S = 0$. Thus b annihilates every column of V and is parallel to a. Primitivity and positivity force $a = b$. □

Lemma 5. *If $n/Q \ge T_d$ and $m \ge 2^{8d^2}$, then (10) holds. With $G = (d!n/Q)^{1/d}$, the maximum ith axis coordinate of $D_n(a)$ lies strictly between $GL_i/2$ and $2GL_i$.*

Proof. The disjoint half-open unit cubes based at the points counted by $F_a(t)$ contain the positive simplex of height t, and are contained in the simplex of height $t + \sum_i a_i$. Therefore

$$\frac{t^d}{d! \prod_i a_i} \le F_a(t) \le \frac{(t + \sum_i a_i)^d}{d! \prod_i a_i} \qquad (t \ge 0).$$

Use $F_a(h-1) < n \le F_a(h)$ and (8) to get

$$\frac{9}{10}G - \frac{11d}{10m} < \frac{h}{H_a} < \frac{11}{10}G + \frac{1}{H_a}.$$

Since $m \ge 10d$ and $G \ge 64d^2(K_d + 1)$, this implies $4G/5 < h/H_a < 6G/5$, and also (10). If M_i is the axis maximum, all integer axis points below score h are included and none above it are included. Thus $h/a_i - 2 < M_i \le h/a_i$. Combining these inequalities with (8) gives $GL_i/2 < M_i < 2GL_i$. □

4 Summing aspect ranges

Proof of Theorem 1. The upper bound is (1). For the lower bound, take R maximal with $Q = 4^R \le n/T_d$. For every tuple

$$r_1 + \cdots + r_d = R, \qquad r_i \ge \lceil R/(d+1) \rceil,$$

put $L_i = 4^{r_i}$. When $n \ge n_*(d)$, these tuples satisfy $R \ge R_0(d)$, $\min_i L_i \ge 2^{8d^2}$, and $\max_i L_i \le (\min_i L_i)^2$. Lemmas 3–5 therefore give at least $2^{-d}\rho_d Q^{d-1}$ different n-point cuts per tuple.

The value G in Lemma 5 is common to all tuples. Two distinct tuples differ by a factor of at least four in one L_i. The open observable ranges $(GL_i/2, 2GL_i)$ are disjoint for that coordinate, so no cut is counted twice across tuples.

Put $r = R - d\lceil R/(d+1) \rceil$. The number of tuples is

$$\binom{r+d-1}{d-1} \ge \frac{R^{d-1}}{[2(d+1)]^{d-1}(d-1)!},$$

since $r \ge R/(d+1) - d \ge R/[2(d+1)]$. Also $Q > n/(4T_d)$ and $R \ge (\log_4 n)/2$. For the latter explicit bound, $d^2 \le 4^{d-1}$ gives $T_d \le 4^{2d^2+4d}$, and $R_0(d) \ge 2d^2 + 4d + 2$. Multiplying the counts yields (2). □

5 Consequences and limitations

There is an established geometric interpretation. For a cut D, put $\alpha_D = \sum_{x \in D} x$. For fixed size n, these vectors are distinct and each is a vertex of their convex hull. Indeed, weights strictly cutting off D make $w \cdot \alpha_D$ smaller than $w \cdot \alpha_E$ for every other n-point set E: each replacement point has greater weight than every removed point. Thus α_D is uniquely exposed. Conversely every vertex of the finite hull is one of its generators. The vertex count of the corner-cut polytope is consequently $C_d(n)$. This correspondence is known [1, 5, 9]; in particular, Müller [5, Lemma 2.6] gives the exposed-vertex argument. Our contribution is the quantitative lower count, not the dictionary. For generic experimental designs the same count describes their corner-cut models [9, Definition 7 and Theorem 2].

Proposition 6 (Realization by prime bases)**.** *For every nonempty finite corner cut $D \subset \mathbb{Z}^d_{\ge 0}$ and every $B > 0$, there are distinct primes $p_1, \dots, p_d > B$ such that D is the exponent set of the first $|D|$ smooth numbers on these ordered bases.*

Proof. Write $D = \{x : w \cdot x < 1\}$ strictly, and set

$$g = 1 - \max_{x \in D} w \cdot x > 0, \qquad H = \max_{x \in D} \sum_i x_i.$$

Relabel the coordinates temporarily so that $w_1 \le \cdots \le w_d$. Choose $T > 0$ with $e^{Tw_1} > \max(B, 2)$ and $dH \log 2/T < g$. Bertrand's postulate [6] supplies a prime in each interval

$$2^{i-1} e^{Tw_i} < p_i < 2^i e^{Tw_i}.$$

These intervals are disjoint, so the primes are distinct, including when some weights agree. Put $v_i = \log p_i / T$. Then $0 < v_i - w_i < d \log 2/T$. For $x \in D$, $v \cdot x < w \cdot x + g \le 1$, whereas for every $x \notin D$, $v \cdot x \ge w \cdot x > 1$. Consequently the products for $x \in D$ are precisely those below e^T. Unique factorization makes their order strict. Undoing the coordinate relabeling proves the statement. □

For fixed B, let $S_{d,B}(n)$ count distinct exponent sets of length n obtained over ordered tuples of distinct primes greater than B. Count each set once, however many tuples realize it. The proposition and the converse observation in Section 1 give

$$S_{d,B}(n) = C_d(n) = \Theta_d(n^{d-1}(\log n)^{d-1}),$$

$$\sum_{j=1}^{n} S_{d,B}(j) = \Theta_d\big(n^d(\log n)^{d-1}\big).$$

The realization argument is an elementary use of a classical theorem; it is not a separate novelty claim. Its role is to specify exactly which smooth-number shapes the counting result describes. A single fixed prime tuple gives one prefix of each length.

The pointwise count gives a matching asymptotic order in every fixed dimension and for the established polytope and model dictionaries. It supplies no asymptotic leading constant, no new enumeration algorithm, no uniform claim as d grows with n, and no fixed-prime unranking complexity improvement. The prescribed-size step is Lemma 4; the earlier cumulative construction alone did not give that conclusion.

6 Prior results and reproducibility

The following distinctions delimit the contribution being assessed. Müller [5, Sections 2–5] studies the same polytopes, including their planar vertex order and adjacency. Her enumeration discussion [5, Section 1, p. 324] cites the planar result and Wagner's bounds; her structural theorems do not supply the bounded-cardinality count proved here.

The unrestricted density of rectangular primitive integer matrices is known [7, 8]. The later paper repairs a defect in the earlier density proof. A centered-cube density alone does not assert the uniform estimate for our moving boxes; Lemma 2 gives a direct finite proof. We claim no new minor criterion or unrestricted density formula.

Chen, Feng, Liu and Wu [14, Theorem 1] give finite probability bounds for appending independent rows to a prescribed primitive matrix. Their random entries use a common interval of length at least the maximum prescribed entry. Our boxes have translated intervals of different lengths, with their shortest length potentially much smaller than the largest entry. That theorem is not a direct substitute for Lemma 2. Prime-by-prime rank analysis and finite determinant cutoffs are established methods; our claimed contribution is their use in the multiscale corner-cut count. The additional fixed-size step is the two-patch comparison in Lemma 4.

Finite-grid threshold encodings and planar counts are related background [11, 12]. Their parameters are the ambient grid and, in the latter case, intersections of two halfplanes. The argument here additionally controls the number of included points, restricts the positive intercepts, and sums disjoint aspect ranges. A difference in definitions alone is not evidence of novelty.

The Hilbert zonotope refines state polyhedra and supports polynomial-time computation of universal Gröbner bases in fixed dimension [10, Proposition 3.2 and Theorem 3.5]. Its count is an upper bound for a common refinement, not the lower bounded-cardinality count proved here. Likewise the lattice-subspace covering bounds of Balko, Cibulka and Valtr [13, Theorems 3–5] concern another counting parameter. Neither result is being recast as a new theorem.

Elizalde and Galván [16, Section 8.2] discuss higher-dimensional pyramidal partitions and refer to the corner-cut growth literature. For this draft, the relevant results in their 2025 journal article are provisionally assumed unchanged from arXiv:2312.16353v1, as authorized

for the project. The final-version comparison is recorded separately for later checking. This assumption does not establish priority.

Kalka and Kuhnt [15, Section 2.2, p. 6] explicitly restate Wagner's lower and upper orders for corner cuts, while treating a numerical statistical fan. That passage does not supply a stronger corner-cut counting bound or a proof of our prescribed-size lemma.

Current priority status. The proof establishes the stated inequalities and their strict improvement over the inspected lower bound in (1). The literature audit has not established an unrestricted claim of priority or submission readiness. In particular, the complete 2002 journal text has not been inspected; the proofs were compared with the author's full thesis chapter. The 2007 grid-encoding paper was compared at the abstract level only. The detailed audit accompanies the source at `results/research/corner_cut_growth_audit.json`. This paragraph must be resolved before the draft is described as submission-ready.

Exact checks. The standard-library Python checker `scripts/run_corner_cut_growth_checks.py` uses independent determinants, modular Gaussian elimination and exact rational feasibility. It checks primitive-plane injectivity, the conditional residue bound, a nonprimitive-projection negative control, and a retained degenerate-arrangement fixture with 18 interior vertices and 48 chambers. The prescribed-size checks construct integer Bezout increments and affine-coset reductions, then verify reversed orders on the two patches for 2,280 ordered pairs of distinct normals in dimensions two through six, each at five boundary heights with different residues relative to the witness height. All 11,400 pair–height checks pass. The exact points and scores are saved. The 200 tie-resolution fixtures include 150 partial boundary levels and 50 complete ones. It exhausts stated small boxes in dimensions two through four and uses deterministic samples in dimensions five and six. Six prime-realization fixtures, including equal weights and unused coordinates, are checked by integer products and an independent heap generation of the corresponding smooth-number prefixes. The explicit constants and 352 scale-rounding cases in dimensions two through twelve are checked with rational arithmetic. These computations can falsify steps; they do not prove the uniform theorem or its novelty.

Run, from the repository root,

```
python3 -B scripts/run_corner_cut_growth_checks.py
```

The report records fixtures, witnesses, the exact command and SHA-256 hashes of its sources in `results/research/corner_cut_growth_checks.json`. The proof above is independent of the computation. The other project kernels and public rank/unrank interface are unchanged.

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